PAUL LONDON

Data Scientist | Biotech, Machine Learning & AI Applications
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SUMMARY

Data science professional with 5+ years of molecular biology experience and applied machine learning expertise. Skilled in Python, predictive modeling, AI/LLM techniques, data visualization, and workflow automation. Experienced in building reproducible computational pipelines and delivering actionable, data-driven insights for scientific and business challenges. Passionate about leveraging analytical and programming expertise to solve complex problems in biotech, healthcare, or AI-driven data science.

TECHNICAL SKILLS

- **Data Science, Machine Learning, & AI:** Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), statistical modeling, regression/classification/clustering, feature engineering, hyperparameter tuning, model evaluation, predictive modeling, AI/LLMs, NLP, embeddings, RAG pipelines
- **Data Analysis & Visualization:** Exploratory data analysis (EDA), data cleaning and preprocessing, interactive dashboards, visualization best practices
- **Workflow & Platforms:** Linux, Snakemake, Git/GitHub, command-line tools, workflow automation, reproducible pipelines, Jupyter/Colab
- **Domain Knowledge:** Genomics, Next-Generation Sequencing (NGS), molecular diagnostics, structural protein analysis, laboratory workflows

TECH PROJECTS

Biotech Trend Intelligence | Google AI Agents Intensive Capstone | 12/25 | GitHub

- Developed an AI (Gemini) platform extracting automated trends from biotech RSS feeds.
- Built an interactive dashboard for real-time exploration of trends.
- Applied retrieval, NLP, and data visualization techniques; submitted as a Concierge Agent project.

Machine Learning Classification of Structural Protein Sequences | Drug Discovery Project | 11/25 | GitHub

- Compared NLP, LSTM, and LLM approaches to predict protein functional class from sequence data.
- Engineered sequence features and embeddings to optimize model accuracy.
- Visualized model performance and protein embeddings for actionable drug discovery insights.

NGS Variant Analysis Pipeline | Bioinformatics Project | 08/25 | GitHub

- Developed an in-house GATK-based NGS pipeline in Linux/Python using Snakemake for FASTQ → BAM → VCF → BED processing.
- Leveraged Biopython, Pandas, NumPy, Plotly, GATK, BWA, Samtools, Picard, BEDTools, and BCFtools.
- Demonstrated pipeline automation, reproducibility, and validation using deidentified sequencing reads.

Property Type Prediction Dashboard | Machine Learning Externship Project | 09/25 | GitHub

- Built a predictive model classifying properties as Detached, Attached, or Condo using 10,000+ MLS records.
- Engineered features and optimized LightGBM and other models with Optuna, achieving **96.6%** classification accuracy.
- Created an interactive Streamlit dashboard with user instructions and downloadable results for dynamic exploration of predictions.

EXPERIENCE

Machine Learning Extern | Berkshire Hathaway HomeServices | 09/25 – 10/25 | Remote

- Partnered with a licensed real estate agent to build a predictive property classification model, cleaning and preparing 10,000+ MLS records (tax, property category, condition) for actionable real-world insights.
- Implemented multiple machine learning models (LightGBM, XGBoost, CatBoost, Random Forest) with Optuna after benchmarking with a baseline LazyClassifier, improving real-world property classification consistency across all categories.
- Built and deployed a Streamlit dashboard with downloadable results, enabling stakeholders to explore property predictions and make data-driven decisions.

Molecular Technologist | Elite Bio Reference Laboratory | 04/24 – 08/25 | Tampa, FL

Independent reference laboratory leveraging advanced molecular technologies for genetic testing:

- Performed routine clinical NGS testing and optimized workflow, achieving a **97% sample pass rate**
- Engineered Excel and script automation to streamline sample processing, halving turnaround time
- Led regulatory documentation efforts in startup lab environment; achieved **92% COLA inspection score**
- Optimized genetic test analysis SOPs, resulting in a **75% reduction in QC incidents**
- Trained other personnel in best practices and workflow optimization to reduce errors

Laboratory Manager | SunPath (Medical Diagnostic Laboratories) | 01/20 – 04/24 | Tampa, FL

Independent reference laboratory utilizing heavy automation and robotics for infectious disease testing:

- Oversaw automated molecular diagnostics lab; ensured inspection readiness for CAP inspections with **zero** deficiencies
- Improved lab **workflow efficiency by 25%**, exceeding expected turnaround targets
- **Supervised and trained 15 lab personnel** while troubleshooting robotics and assays, maintaining high throughput and accurate testing

Client-Facing Professional Experience | 2011 – 2020

- Developed transferable skills in attention to detail, professional communication, and workflow reliability in client-facing and customer service roles

Graduate Research Assistant | University of South Florida | 01/10 – 08/11

- Conducted structural and functional studies of penicillin-binding proteins via X-ray crystallography and computational modeling

EDUCATION | CERTIFICATION

Genomic Data Science Specialization | Johns Hopkins University (Coursera) | 06/25 – 09/25

Coursework: Python, Bioconductor, command-line genomic tools, DNA sequencing, variant analysis

Data Science Professional Training Program | TripleTen | 09/24 – 04/25

Project-based training in machine learning, data visualization, and predictive modeling

Clinical Laboratory Supervisor | SU53237 (FL) – MDxT (AAB) | 2020

Licensed by Florida and accredited by AAB/ABOR as a Molecular Diagnostics Supervisor

M.S. Biotechnology | University of South Florida | 2010 | GPA 4.0

Molecular biology, pharmaceutical development, genetic engineering, bioinformatics

B.S. Biochemistry & Molecular Biology | University of Florida | 2008 | GPA 3.98 | Magna Cum Laude

Genetics, structural biology, microscopy, laboratory research